

Why are we doing Fermi problems of cell biology?

Why am I showing you “random” numbers and names in biology?

Is this quantitative modelling?

Living and intelligent systems are frontiers for physics

Statistical mechanics

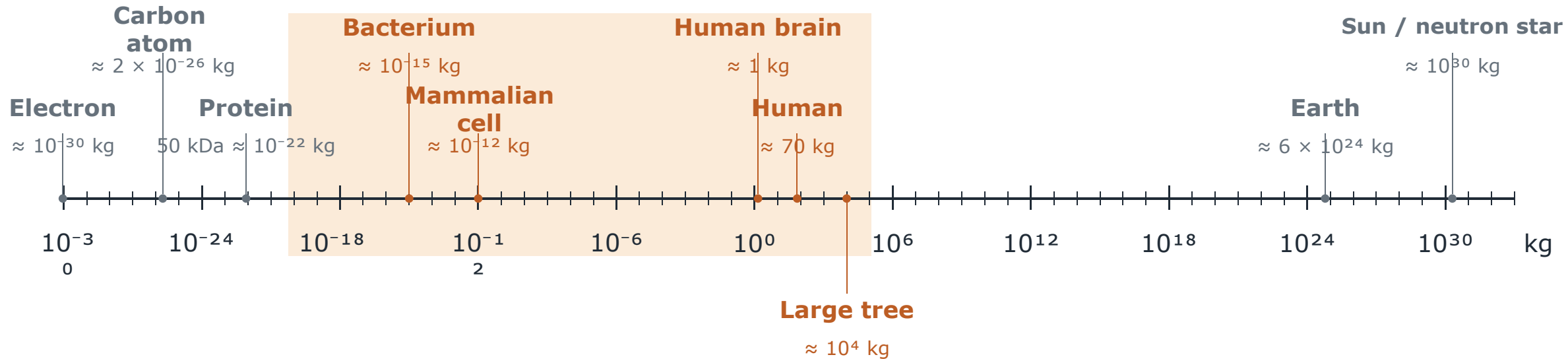
Collective behavior, including living matter

Quantum mechanics

Atoms and molecular interactions

General relativity

Compact stars and strong gravity

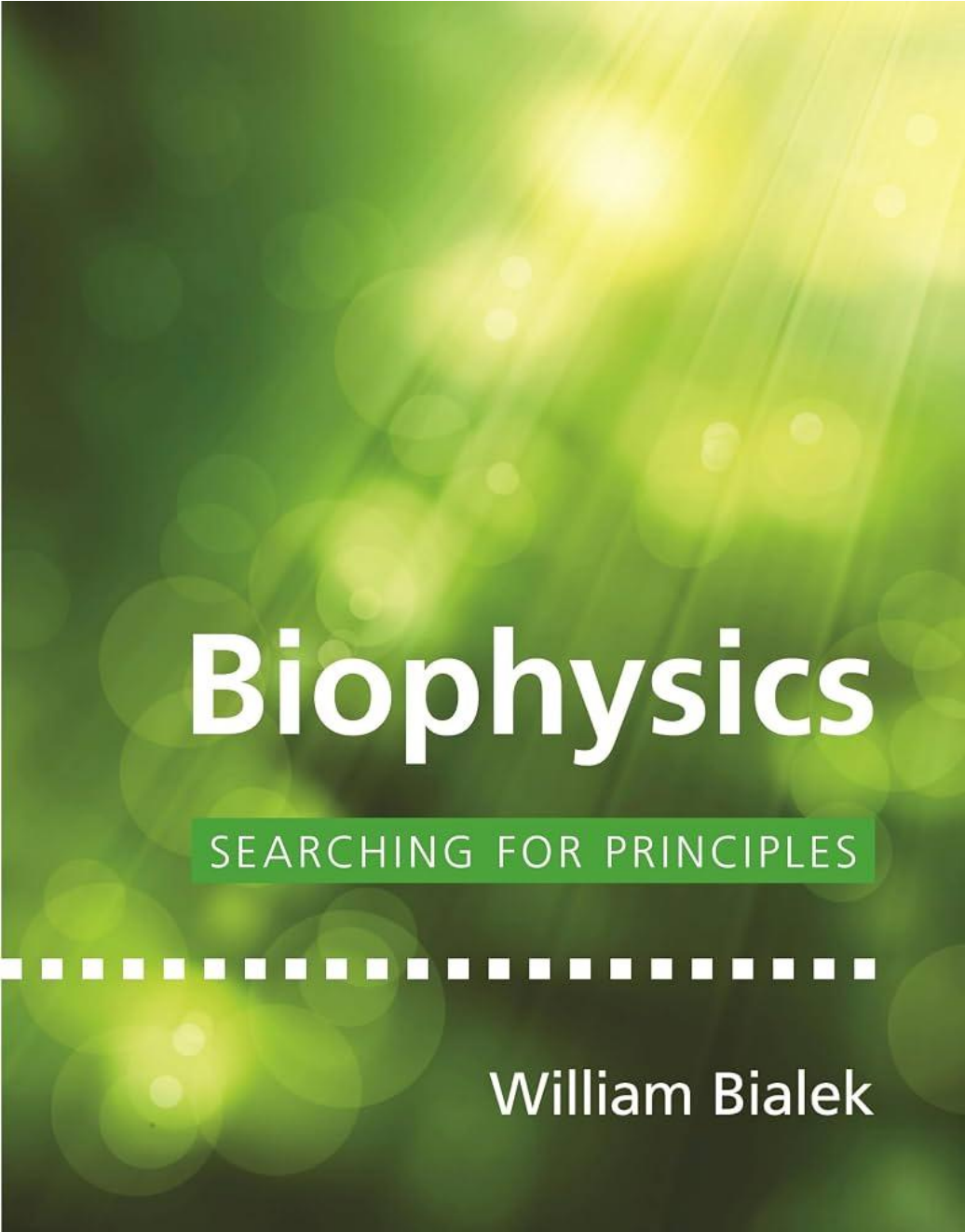


Quantitative biology and complex systems

The frontier you are entering

Masses are approximate. Theory labels mark representative applications, not boundaries of validity.

Living systems obey physical laws. Their organization still demands observation, estimation and model building.



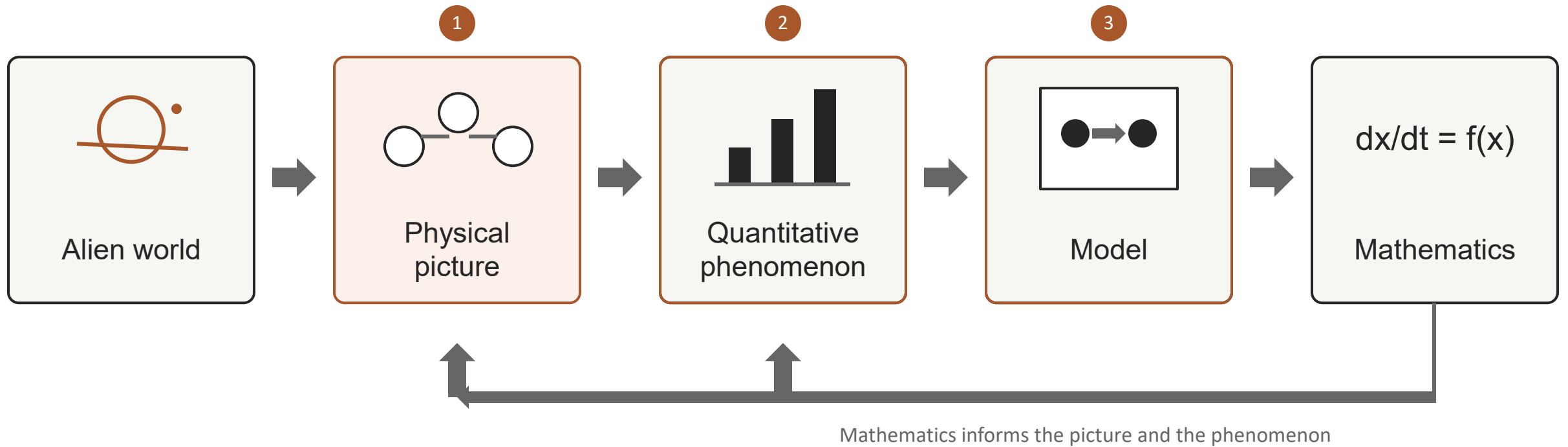
Biophysics

SEARCHING FOR PRINCIPLES

William Bialek

- CIS aims to explain living, intelligent and complex systems quantitatively.
- This requires principles beyond the underlying physical laws. Many remain to be discovered.
- The search requires translating real phenomena and questions into mathematical models.

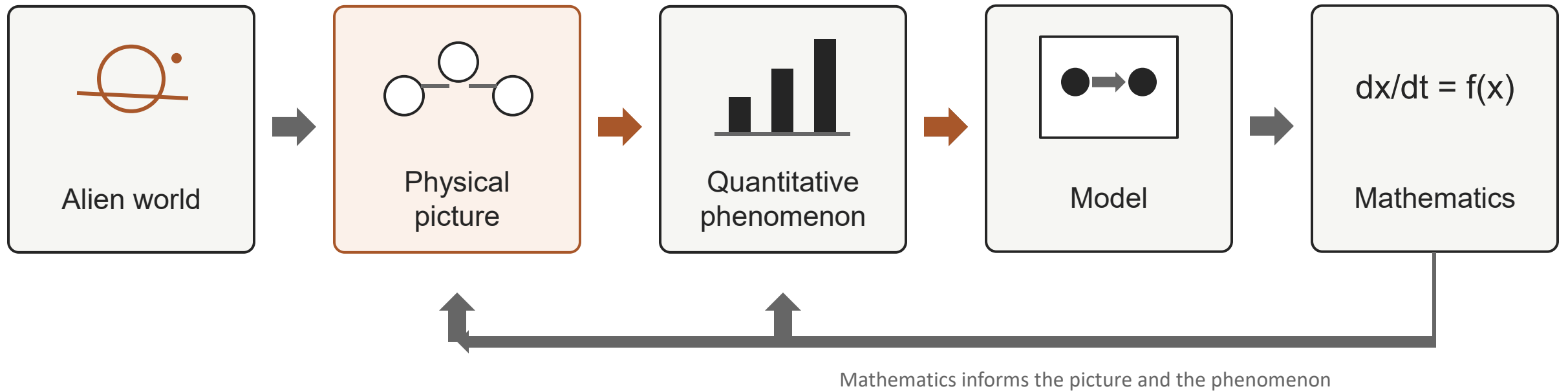
How to build mathematical models



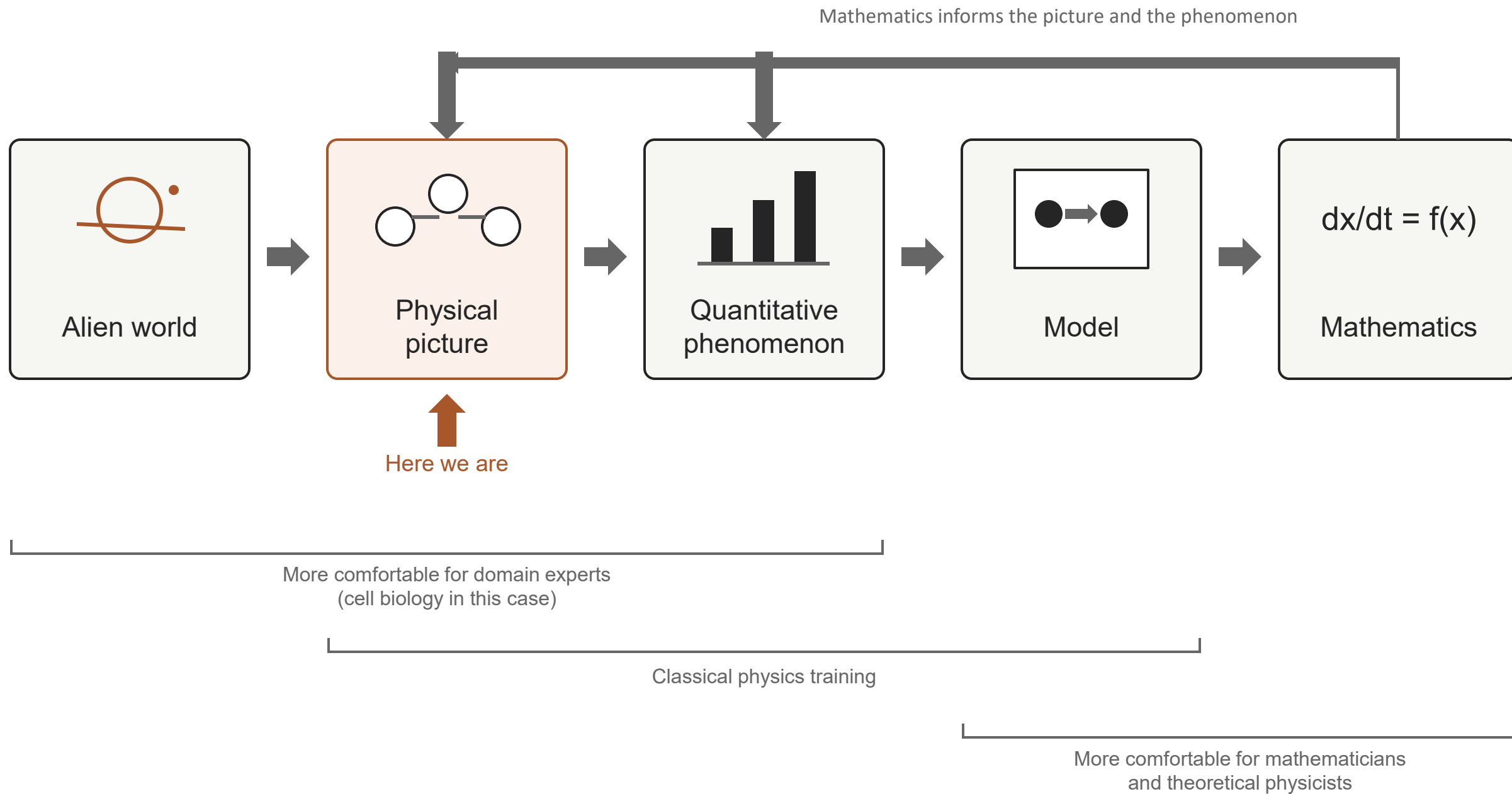
Facts vary across systems; the model-building skill transfers.

In the era of AI, all you have is attention

Critical steps in research



Fermi problems train attention to relevant information and questions



The course is designed for heterogenous backgrounds, but with shared expectations

- Biology: learn the core concepts, look up unfamiliar terms and ask questions. Use the gallery for orientation rather than memorization.
- Mathematics: review needed basics, such as ODEs, as they arise. Class will use real biological examples rather than a separate remediation sequence.
- Everyone: The course expects active information seeking. You are expected to become a professional researcher in a PhD program. Look stuff up or simply ask a question in class.

Where the course goes next

- Shift from numbers to nonlinear dynamics (starting Sept. 15).
- We will turn physical pictures and quantitative phenomena into mathematical models.
- Throughout the course, we will use these models to reveal and test candidate organizing principles.
- You are expected to engage constructively and critically with these principles by testing them against real phenomenon and real data.